7.1 Power Spectral Density Estimation and Spectral Analysis of Noisy Sinusoidal Signals Using Periodogram Method in MATLAB

```
clc; clear; close all

fs=1000; t=(0:999)/fs;
s=sin(2*pi*100*t)+0.5*sin(2*pi*200*t);
n=0.3*randn(size(t));
x=s+n;

figure, plot(x), grid on

title('Signal with Noise')

figure, periodogram(x,[],[],fs)
title('PSD using Periodogram')

figure
X=abs(fft(x));
f=(0:length(X)-1)*fs/length(X);
plot(f(1:500),X(1:500)), grid on
title('Magnitude Spectrum')

figure
bar([mean(s.^2) var(n)])
set(gca,'XTickLabel',{'Signal','Noise'})
title('Power Comparison')
grid on
```

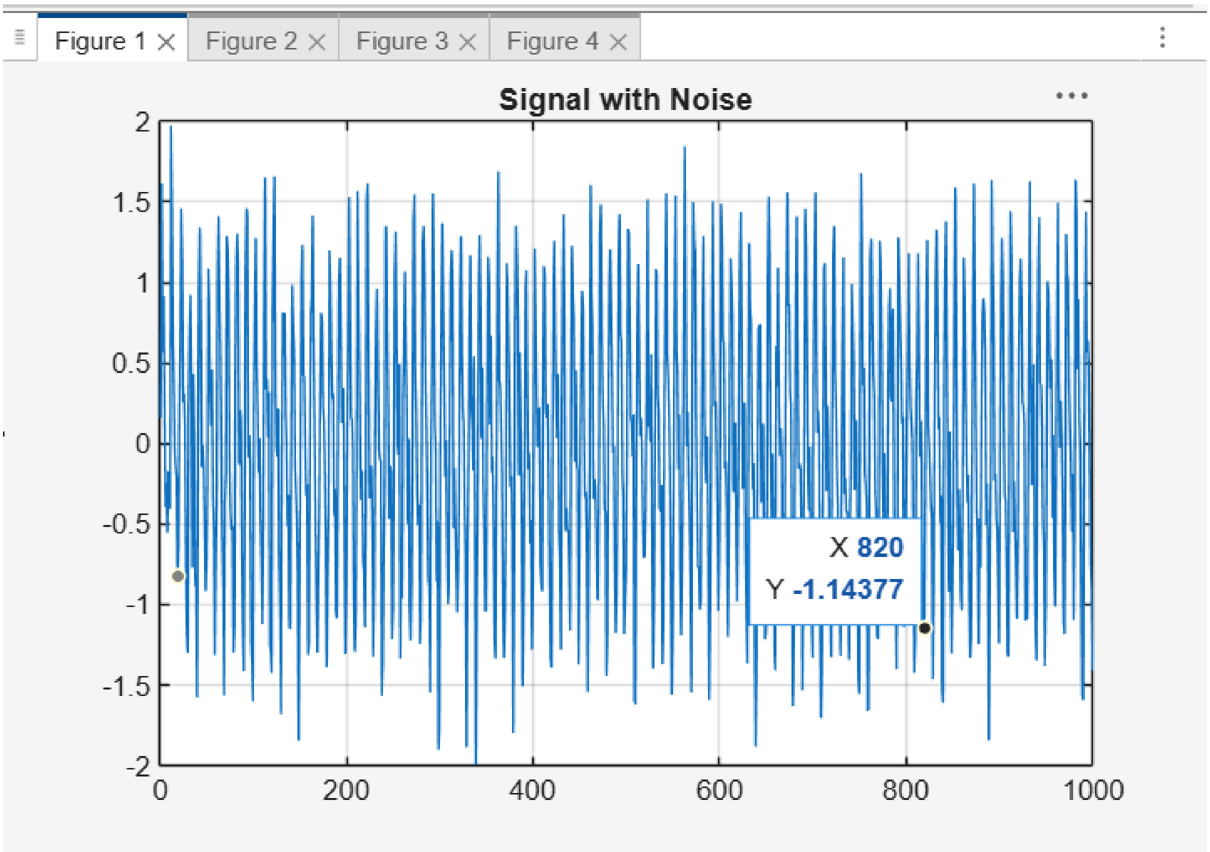
Figure 1
Figure 2
Figure 3
Figure 4
Signal with Noise
X 820
Y -1.14377
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0
200
400
600
800
1000

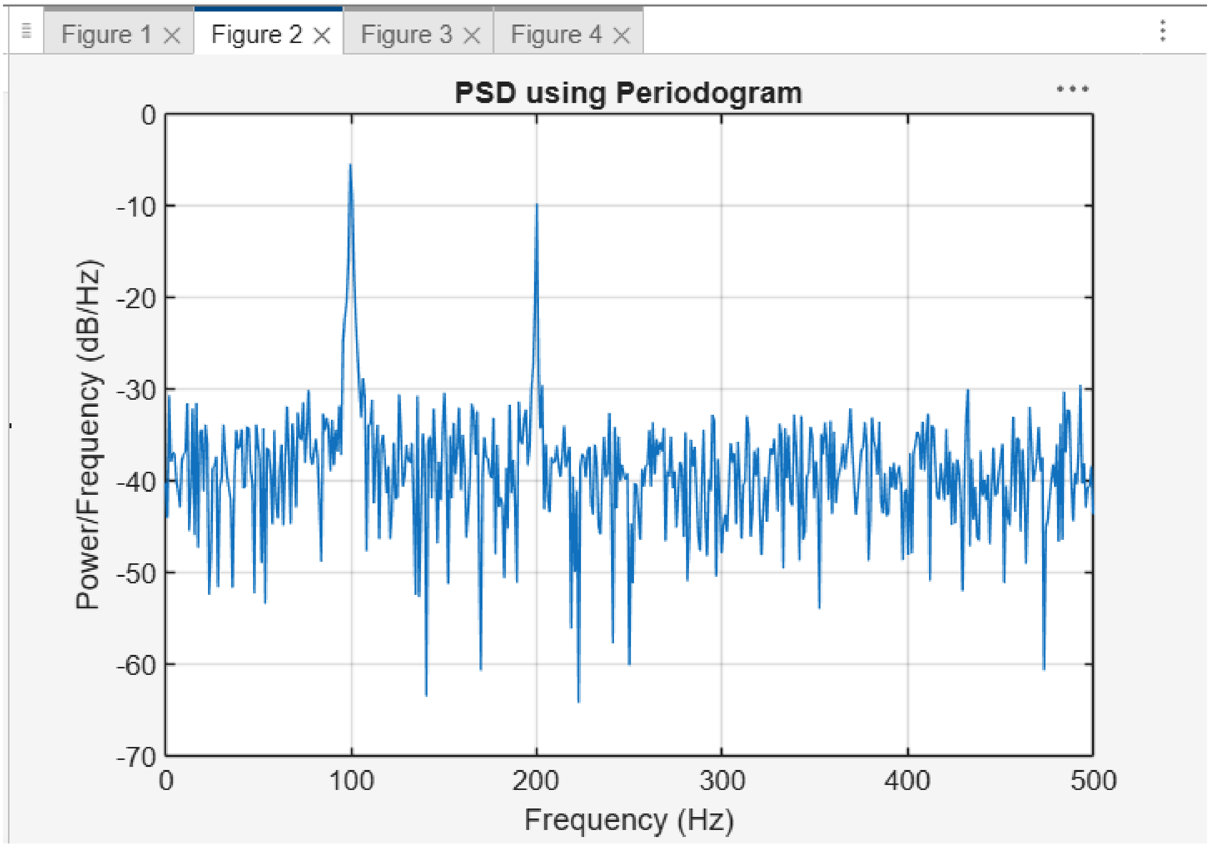
Figure 1
Figure 2
Figure 3
Figure 4
PSD using Periodogram
Power/Frequency (dB/Hz)
0
-10
-20
-30
-40
-50
-60
-70
0
100
200
300
400
500
Frequency (Hz)

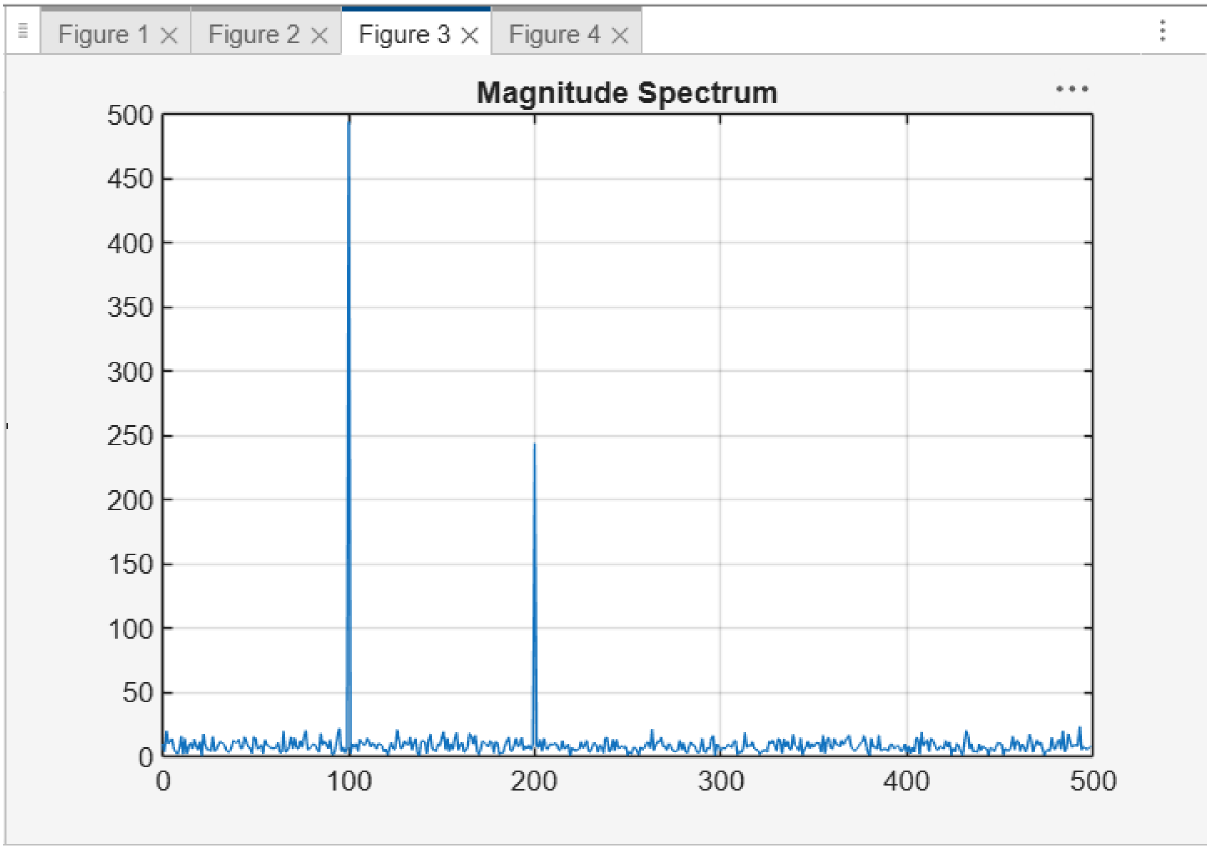
Figure 1
Figure 2
Figure 3
Figure 4
Magnitude Spectrum
500
450
400
350
300
250
200
150
100
50
0
0
100
200
300
400
500

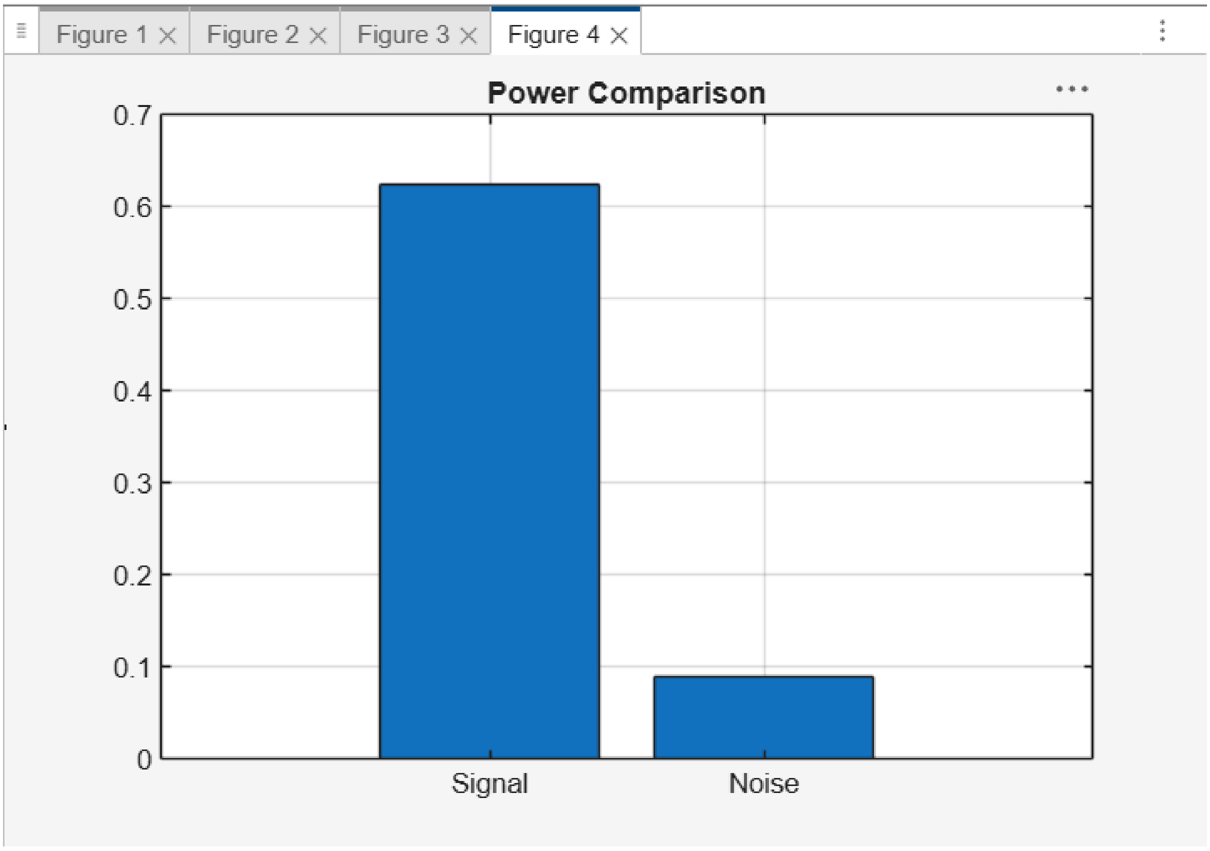
Figure 1
Figure 2
Figure 3
Figure 4
Power Comparison
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
Signal
Noise